

FRIDAY, SEPTEMBER 25

7 Working with data

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This morning

One question, five spreadsheets, and a morning: which countries generated the most revenue, and how do you know.

Where the data is

What an organization actually holds, which system is the record, and why the copies disagree.

Whether to believe it

The joins, columns and blanks that produce a wrong total with no error message.

7 • WORKING WITH DATA

The country revenue report, then the four ways the number comes out wrong.

“We should point AI at our data”

Heard in every company this year
We should point AI at our data.

Which data?

In whose system?

Entered by whom, and
when?

What the surveys say

897

APPLICATIONS IN THE AVERAGE
ENTERPRISE

29%

OF THEM CONNECTED TO
ANYTHING ELSE

7%

OF LEADERS CALL THEIR DATA
READY FOR AI

In the same surveys, 19% of data is called siloed or unusable, and 70% of leaders believe their most valuable insights are inside that 19%.

Where a number actually lives

	HOLDS	REACHABLE BY
The operational system	what is true right now	its own screens
The warehouse	a nightly copy, reshaped for questions	analysts with SQL
A dashboard	four questions somebody chose in 2024	anyone
An extract	one analyst's copy, as of some Tuesday	whoever they mailed it to
A shared drive	the documents none of the above hold	anyone who can find it

Four systems, four meanings of “customer”

IN THIS SYSTEM	A CUSTOMER IS	IDENTIFIED BY
Sales	an organization you might sell to	an account number
Finance	an entity you invoice	a bill-to number
The warehouse	an address you deliver to	a ship-to code
Support	whoever opened the ticket	an email address

Four counts of customers, all correct, none equal.

Why you cannot just query the live system

It is built for writing

Thousands of tiny writes a minute, tuned for that rather than for scanning a year.

Your report competes

A heavy query slows the screen somebody is typing an order into.

It knows one corner

Finance cannot see sales. Neither can see the warehouse.

It holds now

The current address and the current price.
Not last year's.

Three kinds of copy, routinely confused

- 1 The warehouse: a governed copy, refreshed on a schedule, with definitions written down.
- 2 The extract: a file somebody pulled once, for one question, and never refreshed.
- 3 The export: a spreadsheet mailed to someone who now treats it as the source.

Two of the three age silently. The number in the deck was right when it was pulled.

The vocabulary you need for the rest of the day

System of record the one place a fact is authoritative

Silo a store nothing else can reach

Semantics whether two columns with one name mean one thing

Lineage where this number came from, through every hop

Quality blanks, duplicates, and values nobody validates

Every one of these shows up in the next four hours as a reason a total is wrong.

Five files, one question

Northwind: a supplier of specialty foods to restaurants and delis. 500 orders, 93 customers, 77 products, 8 categories, spanning 2012 to 2023.

Small enough to hold in your head, and shaped like a company a thousand times larger. What you are about to do to it is what an analyst does to the real thing.

What you are building

Five spreadsheets in `~/data`: orders, order details, customers, products, categories. One question — which countries generated the most revenue, and how do you know.

Read all five parts before you start. Part D is the point of the exercise. Get there even if you have to shorten Part C.

Part A — describe it first

15 MINUTES

Before any analysis, have it report — with the code shown:

- 1 Every file: row count, column names, data types.
- 2 The date range of the orders, and whether each date column is a date or a string.
- 3 Missing values by column, as counts rather than “a few”.
- 4 Which column joins to which, and whether each join is one-to-one or one-to-many.

Write the join map on paper. You will need it in Part D.

Part B — revenue by country

25 MINUTES

- 1 Compute revenue per order line, then total by the customer's country.
- 2 There are two columns that could be the price. Say in writing which one you used and why.
- 3 Report the row count after every merge and compare it to the row count before.
- 4 Rank the countries and take the top five.

There is no discount column and no freight column in these files. Whatever you report is price times quantity, and the deliverable has to say so.

Part C — a deliverable

20 MINUTES

Option 1

An Excel workbook: the country table on one sheet, a chart, and a cell stating the revenue definition in words.

Option 2

A five-slide deck: the question, the top five, the chart, the definition, and what you checked.

Either one states the revenue definition on its face. A number without its definition cannot be reconciled against anyone else's.

Part D — verify it

25 MINUTES

- 1 Recompute the total by a second route — a different tool, a different order of operations, or straight from the raw files with no merges at all.
- 2 Reconcile the two to the cent.
- 3 If they disagree, find the cause before changing either one.
- 4 List three things that could have made your number wrong, and how you ruled each one out.

If the second route reproduces the first to the cent, say so on the deliverable.

Part E — if you finish

REMAINING TIME

- Break revenue out by month and by category
- Find a month that does not follow the months around it
- Say what in the data produces it, and show the rows

Have open and ready to share:

- Your number, and the definition behind it
- Your chart
- Your second route, and what it gave
- The three things that could have gone wrong

When you are done

WHAT WENT WRONG

Four failures that raise no error

The wrong join

Rows that did not match are dropped. The total is smaller and nothing says so.

The wrong column

Two files use one name for two different things, and the merge keeps both.

The unstated assumption

What counts as revenue, which rows are excluded, which dates are in scope.

Invention

A number in prose that no line of code produced.

What is actually in these files

FILE	ROWS	
Orders	500	July 2012 to October 2023
Order details	18,976	38 lines per order on average
Customers	93	in 21 countries
Products	77	in 8 categories

Every order line matches a product, an order, and a customer. No merge in this analysis drops a row — which is why you have to check rather than assume.

Two columns named UnitPrice

Order details carries the price charged. Products carries the list price. Merge them and you get `UnitPrice_x` and `UnitPrice_y`.

PRICED AT	TOTAL REVENUE
The price charged	\$13,963,378.65
The list price	\$13,966,340.55

The two answers differ by \$2,962 on \$14 million. A sanity check on the magnitude passes either way, so nothing about the wrong choice looks wrong.

The rows with no country

2

CUSTOMERS WITH A BLANK
COUNTRY

482

ORDER LINES BEHIND THEM

\$373,609

REVENUE IN THOSE LINES

Grouping by country drops blank keys by default. The country table still adds up, to a total 2.7% below the real one, and no country is wrong.

One-to-many

Merge orders into order details and every order appears once per line — on average 38 times.

Sum anything defined per order after that merge and you have counted it 38 times. On these files the multiplier is large enough to notice. On files where the ratio is 1.2, it is not.

The check is one line: row count before the merge, row count after.

Dates

- `03/04/2023` is March 4 or April 3 depending on who wrote the file
- A date column read as text sorts `10/01` before `2/01`
- These orders carry a time as well as a date, so a filter on equality matches nothing
- A date read out of Excel can arrive as a serial number

Ask which columns are date types and which are strings, before grouping by month.

Missing values

HOW IT IS WRITTEN	WHAT A NAIVE COUNT DOES
An empty cell	usually skipped
0	included in the average
NA, N/A, -, unknown	makes the whole column text
A sentinel such as -999	included, silently

A numeric column that is text because of four N/A cells concatenates where you expected a sum.

- 1 The same question by a different route through the data.
- 2 The same question in a different tool — SQL against the same files, or a pivot table.
- 3 The same question in a fresh conversation that has not seen the first answer.

Each route drops different rows and rounds at different points. Agreement across them narrows what can still be wrong.

Recompute another way

Make it read its own code

Prompt

Here is code and the answer it produced. Ignore the answer. Read the code and tell me what it actually computes – including what it drops and what it assumes.

A fresh conversation reading code has nothing invested in the number already on the screen.

Find the bug

EXERCISE 9 · 40 MINUTES

- 1 `~/data/planted/country_report.py` produces a country table. Its total is wrong.
- 2 Read it first and predict where the number goes wrong. Do not run it yet.
- 3 Now run it, find the actual causes, and say how much each one moves the total.
- 4 Fix them and reconcile against your own Part B number.

There is more than one defect, and they move the total in different directions.

Today the answer was a number you checked. Tomorrow it becomes a document somebody else opens, and then an application they use without you.

Tomorrow